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Working Group V - Isolated Galaxies and Subgrid Physics

[Written by James Wadsley based on the discussion during the 2nd AGORA Workshop, Aug. 16–18, 2013]

The isolated galaxy project has 3 facets.

1) Calibrate so that AGORA is more standardized than other comparison projects.

This may not be publishable on its own merits and is completely open ended in principle. The obvious starting point would be SFR -- specifically sitting on the Kennicutt relation/Galaxy Main sequence. Thus SFR would be probably a two parameter fit -- SFR and also SFR for a given gas content/surface density.

There was a consensus that we would suggest standard parameters (e.g. thresholds, SF rate coefficients, feedback amounts and prescriptions) but each group would not want to be too constrained because they want to get a "good" cosmological galaxy. For example, many gadget groups are only different in how they do winds. If we prescribe the wind properties they will all be the same. Also -- a certain performance on a low z MW analogue is not going to say anything on how well the overall SF history, final stellar mass or other structure properties are going to look. Therefore we decided not to require specific outcomes with respect to outflows, gas properties, etc... for the calibration.

Thus the basic calibration suggestion is that each group should aim to fit two parameters with respect to star formation on the isolated MW -- such as average SFR for 0.5–1 Gyr and SFR for a certain gas content but we would otherwise let groups choose what to do. The goal would be to get the same SFR at all resolutions (with different parameters if necessary). Every group should use the standard AGORA physics (cooling and so on). We anticipate that different groups will need to use different parameters (hopefully close to the standard suggestions) to achieve a basic calibration for $z=0$.

Calibration steps:

- 1) Implement common physics (in all codes)
- 2) Work out what SFR is expected for the standard isolated model
- 3) Propose a standard set of SF/FB parameters that do ok for a couple of codes and seem physically well motivated
- 4) Each code group independently determines a set of parameters, close to the standard set, that can match the SFR expectation on the isolated models using the feedback that group will use for the cosmological runs.

2) Explore the origin of code differences in what it takes to get similar results on an isolated (non-cosmological) galaxy.

We would explore SF recipes, feedback and parameter choices and see how easily different hydro methods can agree or mimic each other. We would develop points of comparison: SF rates, clustering of star formation, phase diagrams, temperatures, density PDF's, velocity dispersions, outflows, fountains, galactic structure ...

For this project we could actively try to use the same parameters as much as possible and explore the origins of the differences. For example, we would try to keep the same thresholds, feedback energies and start with something like a SN feedback with delayed cooling. We would use some simple tests to ensure the implementations are comparable. This project can progress in parallel with the cosmological AGORA comparison and should be publishable on its own merits.

3) Explain why the AGORA galaxies are different.

The cosmological galaxies AGORA makes will be very different. We may be able to use the isolated galaxies to understand why. For example -- we can look at which codes create strong winds and so on. We can also remap the isolated galaxies to higher redshift analogues (where most SF and feedback occur) by dialing up the gas content and changing some structural parameters and explore how codes behave differently then. This would give us more insight into code differences (probably) than the low z case. This project would be a natural follow-up once we get some cosmological runs in place.

Working Group V - Blog

Untitled Post This is the first blog post for the Working Group V – Isolated Galaxies and Subgrid Physics.

Posted Oct 4, 2012, 12:40 AM by Ji-hoon Kim

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